

According to some experts, it would be a good choice to have very little functionality other than just transferring data packets within different modules on the base module. This would keep the basic cost down and ensure enough space for enhancing modular functionality.

The open hardware platform comes with a side effect as well. The device could crash or stop functioning on connecting a sub-standard module designed by some other manufacturer. So, an important consideration while designing the base module would be to include enough safeguards for thermal and power management built on to it. This would secure the device from any functional hazard, if any bad or sub-standard module is connected, thus ensuring normal functioning of the device after the removal of such module.

Interfacing

Connectors between the main base unit and the other modules are also a big performance challenge. If we see integrated designs, these are set up for high performance. Connectivity between different units is highly optimised, but breaking this up and building external connectors that operate at similar performance levels would be a huge challenge.

Any external connector connecting different modules would give the device a 'Lego' kind of look, while at the same time making it bulkier. Moreover, such a modular system would consume more power, since hot swappable and standard connectors would now replace the more fine-tuned custom connections.

Will connectors be the knell of doom?

Well we have alternatives; the device doesn't have to be connector based in the first place. Today, we have architectures which do not require any connectors, by having their electrical interfaces defined in the mechanicals of the device itself. This way the PCB of any module can be connected with

What about the drivers?

Swapping of modules in smartphones will include software update challenges as well. The option of self-discovery seems to work fine in the PC world where it automatically figures out that there is a new thing connected and the processor starts downloading the drivers or seeking for the drivers, but there will be some considerations while implementing modularity in smartphones:

1. Updating the driver software, if you are implementing 'hot swapping' of modules where a module can be replaced without switching off the device.
2. Reading the modular drivers at the time of power-up could accompany a side effect in the form of delay in boot-up time.
3. What about the CPU? Moving from one CPU to another will also require updated drivers for all the modules attached. Where can we get those?

“I think efforts around the base architecture are more important than any specific concept. People should be more focused on trying to standardise the base module and the high-speed interface to connect these other modules. That will go a long way in making modular phone concept a reality” —
Satish Mugulavalli, co-founder & chief architect, Verismo Networks

the PCB of the other modules without the standard connectors. Or the casings of modules could have embedded electrical interconnects to connect the modules.

Nate Srinath, founder-director, Inxee, explains, “If we look at the phones today, we are moving towards thinner and thinner architectures, so most connectors available are big for these. We have to look for very fine customised, thin connectors which are very durable and can make multiple modules connect. Some architecture doesn't have any connectors but the electrical interfaces are defined in the mechanicals itself. If we can dream, there are so many ways to do it, but the bottom line is somebody has to standardise it and then make everybody adhere to it.”

Let's talk solutions

A plausible solution could be electro-permanent magnets. Electro-permanent magnets are passive in both on and off states and can be used to transfer data as well as power. These require a voltage for transition between two states and their off-state magnitude is lower. But when turned on, their magnitude increases many

folds, which is good enough to hold different modules. Also, eliminating any external connector, these connections would allow the device to be slim and maintain its aesthetic looks.

Another contact-less power and communication interface technology which might help is capacitive media converter. The technology is still in development phase where coupling occurs by capacitance at higher frequencies and inductance at lower frequencies. Experts believe that if connectors are avoided, it could also help in reducing the additional industrial design overheads related to increased volume and weight.

But one needs to carefully consider contact and contact-less options while going for the connections between modules. Ashwin Ramachandra, VP and head, ErND Practice, Sasken Communication Technologies, explains, “The moment you look at contact-less options, a new problem immediately comes up. Integrating issues with respect to signal, power and bandwidth will definitely be there. But for the areas like when you are connecting the processor with memory you can't afford to overlook these issues, you need a fairly strong connector or a